Arista MAR-ACC-30 Major Optical Transport Failure (SFP/XFP Degradation, Fiber Cuts, Rx/Tx Power Loss) Resolution SOP

Node: MAR-ACC-30 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8760

Description

This failure domain occurs on the MAR-ACC-30 node in the Marathahalli Core site due to the inherent vulnerabilities of the Arista Access role's Optical Transport mechanism. The Optical Transport Failure is triggered by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which are critical components of the Access role's data plane. The root cause of this failure domain is the susceptibility of the SFP/XFP interfaces to degradation or damage, leading to a loss of signal strength or complete signal loss. This, in turn, causes a major optical transport failure, resulting in the ALARM_8760 alarm being raised. The underlying protocol mechanics involve the Optical Transport protocol, which is responsible for transmitting data between nodes over fiber-optic cables. The hardware mechanics involve the SFP/XFP interfaces, which are responsible for transmitting and receiving optical signals.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and all necessary personnel have been dispatched.
- Check the redundant link status to ensure that the network is still operational.
- Verify that all necessary safety protocols have been followed and that the site is safe for maintenance.

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cuts	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status: Up or Down, Description: This command checks the status of the interface and identifies whether it is up or down. An up interface indicates that the SFP/XFP is functioning correctly, while a down interface indicates a potential issue with the SFP/XFP or the fiber connection.

Command: show interface optics, Expected Output: Optics status: Good or Bad, Description: This command checks the status of the optics on the interface and identifies whether they are functioning correctly. A good optics status indicates that the optics are functioning correctly, while a bad optics status indicates a potential issue with the optics.
```

```
Command: show interface transceiver, Expected Output: Transceiver status: Good or Bad, Description: This command checks the status of the transceiver on the interface and identifies whether it is functioning correctly. A good transceiver status indicates that the transceiver is functioning correctly, while a bad transceiver status indicates a potential issue with the transceiver.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace interface SFP-XFP, Description: This step involves replacing the SFP/XFP on the interface to resolve the issue. The new SFP/XFP should be properly seated and secured to ensure proper connection.

Step 2: Step: Verify the fiber connection, Command: verify fiber connection, Description: This step involves verifying the fiber connection to ensure that it is properly connected and secure. The fiber connection should be checked for any signs of damage or wear.

Step 3: Step: Reset the interface, Command: reset interface, Description: This step involves resetting the interface to its default state. This will clear any configuration and bring the interface back to its default state.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status: Up, Description: This command checks the status of the interface and verifies that it is up and functioning correctly.

Command: show interface optics, Expected Output: Optics status: Good, Description: This command checks the status of the optics on the interface and verifies that they are functioning correctly.

Command: show interface transceiver, Expected Output: Transceiver status: Good, Description: This command checks the status of the transceiver on the interface and verifies that it is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore interface SFP-XFP, Description: This step involves restoring the original SFP/XFP on the interface to its original state.

- Step: Reconfigure the interface, Command: reconfigure interface, Description: This step involves reconfiguring the interface to its original state. This will restore any configuration that was previously removed.